



DW_bc_5

Boundary Scan Cell Type BC_5

Version, STAR and Download Information: [IP Directory](#)

Features and Benefits

- IEEE Standard 1149.1 compliant
- Synchronous or asynchronous scan cells with respect to t_{ck}
- Supports the standard instructions: EXTEST, SAMPLE/PRELOAD, and BYPASS
- Supports the optional instructions INTEST, RUNBIST, CLAMP, and HIGHZ

Description

DW_bc_5 is a boundary scan cell used to control the output enable for a three-state output buffer when a signal received from an IC input pin is used only as an output enable. DW_bc_5 combines the functions of an input cell and an output cell. The Boundary Scan Description Language (BSDL) description of this cell is of type BC_5 described in the BSDL package STD_1149_1_1990.

The DW_bc_5 cell may be synchronous or asynchronous with respect to t_{ck} (Test Clock system pin), depending on the port connections.

Revision History

—	data_in	data_out	—
—	si	so	—
—	intest		
—	mode		
—	shift_dr		
—	capture_en		
—	update_en		
—	update_clk		
—	capture_clk		

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
capture_clk	1 bit	Input	Clocks data into the capture stage
update_clk	1 bit	Input	Clocks data into the update stage
capture_en	1 bit	Input	Enable for data clocked into the capture stage, active low
update_en	1 bit	Input	Enable for data clocked into the update stage, active high
shift_dr	1 bit	Input	Enables the boundary scan chain to shift data one stage toward its serial output (tdo)
mode	1 bit	Input	Determines whether data_out is controlled by the boundary scan cell or by the data_in signal
intest	1 bit	Input	INTEST instruction signal
si	1 bit	Input	Serial path from the previous boundary scan cell
data_in	1 bit	Input	Input data from system input pin

Table 1-1 Pin Description (Continued)

Pin Name	Width	Direction	Function
data_out	1 bit	Output	Output data
so	1 bit	Output	Serial path to the next boundary scan cell

Table 1-2 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare or Test-IEEE-STD-1149-1

The `mode` signal gives the Test Access Port (TAP) instructions control of the boundary scan cell. [Table 1-3](#) lists the required values of the `mode` signal for each of the TAP instructions that DW_bc_5 supports.

Table 1-3 Mode Signal Generation for DW_bc_5

Instruction	Mode for Output Cell
EXTEST	1
SAMPLE/PRELOAD	0
INTEST	1 ^a
CLAMP	1
RUNBIST	1 ^a
BYPASS	0

- a. If you do not want these instructions to drive the output pins with pre-loaded data held in the boundary scan register, then these instructions are not needed to determine the state of the `mode` signal. Instead, the instruction must be added to the output enable logic to force every system output pin to an inactive drive state.

[Table 1-4](#) lists the connections for asynchronous boundary scan chains.

Table 1-4 Simulation Models

Model	Function
DW04.DW_BC_5_CFG_SIM	Design unit name for VHDL simulation
dw/dw04/src/DW_bc_5_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_bc_5.v	Verilog simulation model source code

[Table 1-5](#) lists the connections for synchronous boundary scan chains.

Table 1-5 Port Connections for Asynchronous Boundary Scan Chains

DW_bc_5 Port Name	Connection
capture_clk	clock_dr from TAP controller
update_clk	update_dr from TAP controller
capture_en	Logic zero
update_en	Logic one
shift_dr	shift_dr from TAP controller
mode	Mode generation logic
si	so from previous boundary scan cell
intest	INTEST signal from the instruction decoder
data_in	System input pin for input cells or IC output logic for output cells
data_out	IC input logic for input cells or system output pin for output cells
so	si of next boundary scan cell

Table 1-6 Port Connections for Synchronous Boundary Scan Chains

DW_bc_5 Port Name	Connection
capture_clk	tck from system pin
update_clk	tck_n from system pin
capture_en	sync_capture_en from TAP controller
update_en	sync_update_dr from TAP controller
shift_dr	shift_dr from TAP controller
mode	Mode generation logic
si	so from previous boundary scan cell
intest	INTEST signal from the instruction decoder
data_in	System input pin for input cells or IC output logic for output cells
data_out	IC input logic for input cells or system output pin for output cells
so	si of next boundary scan cell

Related Topics

- [Application Specific – JTAG Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW_bc_5_inst is
    port (inst_capture_clk : in std_logic;    inst_update_clk  : in std_logic;
          inst_capture_en  : in std_logic;    inst_update_en   : in std_logic;
          inst_shift_dr    : in std_logic;    inst_mode        : in std_logic;
          inst_intest      : in std_logic;    inst_si          : in std_logic;
          inst_data_in     : in std_logic;    data_out_inst    : out std_logic;
          so_inst          : out std_logic );
end DW_bc_5_inst;

architecture inst of DW_bc_5_inst is
begin

    -- Instance of DW_bc_5
    U1 : DW_bc_5
        port map (capture_clk => inst_capture_clk,
                  update_clk => inst_update_clk,   capture_en => inst_capture_en,
                  update_en => inst_update_en,    shift_dr => inst_shift_dr,
                  mode => inst_mode,   intest => inst_intest,   si => inst_si,
                  data_in => inst_data_in,   data_out => data_out_inst,
                  so => so_inst );

end inst;

-- pragma translate_off
configuration DW_bc_5_inst_cfg_inst of DW_bc_5_inst is
    for inst
    end for; -- inst
end DW_bc_5_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_bc_5_inst(inst_capture_clk, inst_update_clk, inst_capture_en,
                    inst_update_en, inst_shift_dr, inst_mode, inst_intest,
                    inst_si, inst_data_in, data_out_inst, so_inst );

    input inst_capture_clk;
    input inst_update_clk;
    input inst_capture_en;
    input inst_update_en;
    input inst_shift_dr;
    input inst_mode;
    input inst_intest;
    input inst_si;
    input inst_data_in;
    output data_out_inst;
    output so_inst;

    // Instance of DW_bc_5
DW_bc_5
    U1 (.capture_clk(inst_capture_clk), .update_clk(inst_update_clk),
        .capture_en(inst_capture_en), .update_en(inst_update_en),
        .shift_dr(inst_shift_dr), .mode(inst_mode),
        .intest(inst_intest), .si(inst_si), .data_in(inst_data_in),
        .data_out(data_out_inst), .so(so_inst) );
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 4 ■ Added this Revision History table and the document links on this page

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